

z/OS for Infrastructure Engineers Using Ansible

INTRODUCTION AND CLASS LOGISTICS





LOGISTICS



Class Hours:

- Instructor will provide class start and end times.
- There will be regular breaks in class.



Telecommunication:

- Turn off or set electronic devices to silent (not vibrate)
- Reading or attending to devices can be distracting to other students
- Try to delay until breaks or after class

Miscellaneous:

- Courseware
- Bathroom
- Fire drills

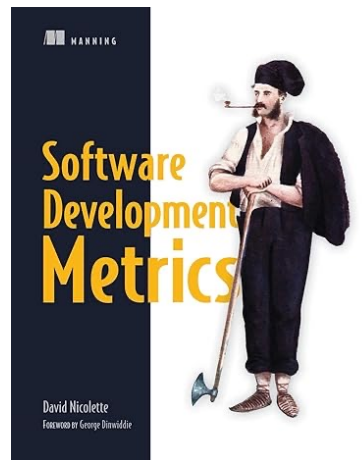
DAVE NICOLETTE

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Technical Coach

Speaker / Author

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Unit Testing and
Test-Driven Development
Self-Check



Dave Nicolette



What Is Professionalism?



Dave Nicolette

Choosing an
Agile Scaling Framework



A handbook for decision-making

Dave Nicolette

INTRODUCE YOURSELF

Time to introduce yourself:

- Name
- What is your role in the organization
- Indicate software development experience



LABS



A significant portion of the class time will be spent in hands-on work.

We will use contemporary collaborative methods such as Ensemble Programming and Pair Programming.

Everyone is expected to participate and to ask for and offer help. It is not a competition.

SCOPE AND OBJECTIVES OF THE TRAINING

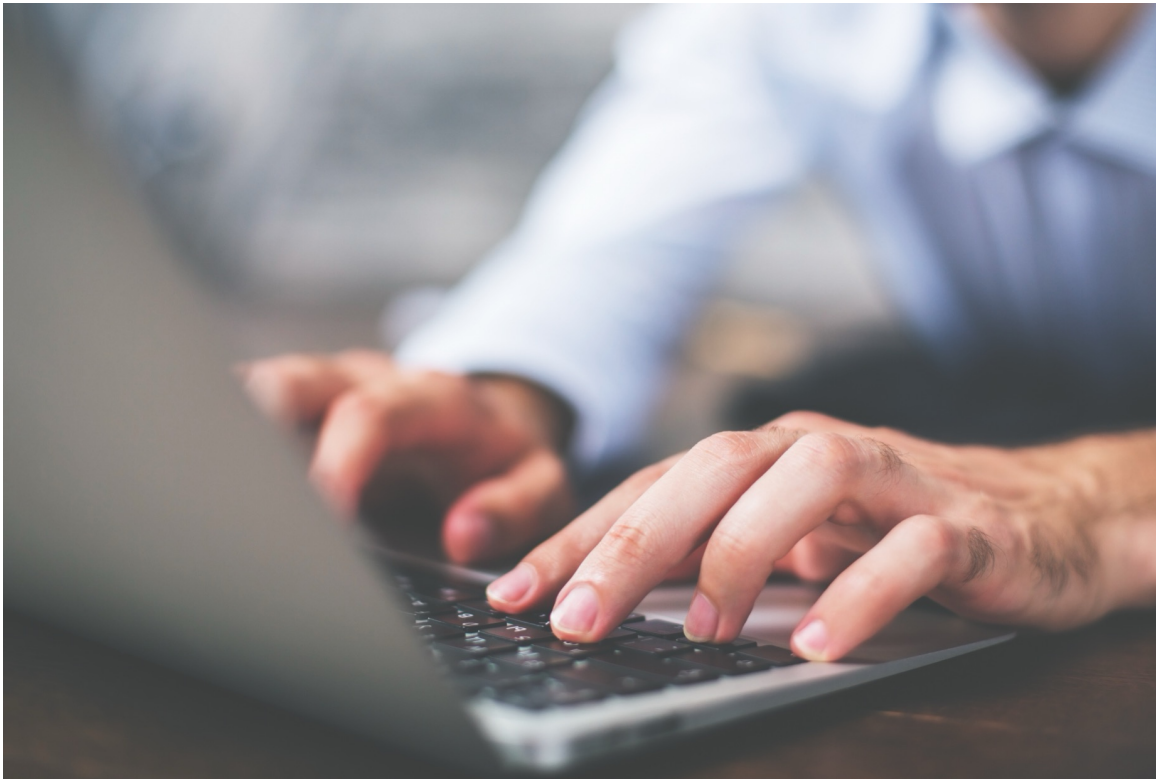
This week-long segment of the bootcamp aims to get you started with using Ansible to manage z/OS instances.

The key skill areas to cover are:

- Some z/OS basics
- TSO and ISPF (Time-Sharing Option)
- JCL (Job Control Language)
- `ibm_zos_core` collection
- `zos_data_set` module
- `zos_tso_command` module
- Additional selected modules



RESOURCES FOR HANDS-ON WORK



We'll take a hands-on approach throughout the training. To enable that, you'll need access to three resources:

- The repository containing samples and data files.
- A 3270 terminal emulator installed locally on your system or accessible to you on your network.
- A user ID and password for the z/OS lab system
- A public key for accessing the Ansible control node

The expectation is that you have received the URL for the repository and credentials for the lab system and you have access to a 3270 emulator. Best case, you have tried these things already to ensure they are functional.

Let's make sure you have everything in place and you can access what you will need to complete the work.

COURSE REPOSITORY

The course repository contains instructions, data files, and starter code for lab work. You should have been given the URL in advance, but if not the instructor will share it.

You can fork the repo and keep your work for future reference.



COURSE LAB ENVIRONMENT

You



ssh

Ansible Control Node



**z/OS Managed
Node**

ssh



3270 terminal emulation

The URL and public key to access the Control Node and the URL and userid/password to access the z/OS environment were provided to you.

ACCESSING THE Z/OS ENVIRONMENT

Main Page

[Main Page](#) [Discussion](#)

[Read](#) [View source](#) [View history](#) [Tools](#)

Welcome to The x3270 Wiki

This Wiki contains documentation for the x3270 family of IBM 3270 terminal emulators.

Navigating the Wiki

The x3270 Wiki is organized in two ways: Hierarchically and by categories.

Hierarchical pages have a slash (/) in their names. The pages under each specific emulator are generally organized in a hierarchy. For example, the page that describes how [c3270](#) keymaps work is called [c3270/keymap](#).

Pages can also belong to one or more **categories**. Typical categories are [command-line options](#) and [resources](#). There are also **sub-categories**, for example, [c3270 command-line options](#). The list of all categories is [\(here\)](#).

The Wiki makes extensive use of hyperlinks. Wherever possible, a term that is defined elsewhere in the Wiki is always represented by a hyperlink.

x3270 variants

Interactive emulators

- [x3270](#) is the original version. It is a graphical interactive 3270 emulator that runs on Linux, Solaris, macOS (with [XQuartz](#) installed), FreeBSD and other [POSIX](#)-based operating systems.
- [wx3270](#) is a modern interactive graphical 3270 emulator for Microsoft Windows.
- [c3270](#) is an interactive 3270 emulator that runs on a character-mode terminal on [POSIX](#) systems. A character-mode terminal can be an emulator such as xterm, gnome-terminal, kterm, a macOS Terminal window, or a Linux console, or a real hardware terminal such as a [DEC VT100](#).
- [wc3270](#) is an interactive 3270 emulator that runs in a console window on Microsoft Windows.



wx3270 main window

For historical reasons, the basic way to access a z/OS system is through an IBM 3270 terminal. Physical terminals are no longer used. People use 3270 emulation software to interact with z/OS.

If you have not already set up a 3270 terminal emulator, please take a few minutes to install one now. You will not be able to complete any lab assignments related to the z/OS topic without one.

The most popular free 3270 emulator is x3270. See the wiki for a version that runs on your local system:

https://x3270.miraheze.org/wiki/Main_Page

START YOUR TERMINAL EMULATOR

Your terminal emulator may be different from mine. I'm using a command-line wrapper around the open source emulator, x3270. To connect to the mainframe, I enter the name of the wrapper program followed by the URL and port that was assigned to us. If you are using a different terminal emulator, follow the instructions for that product to connect to the mainframe.

```
> c3270 67.217.62.83:23
```

THE FIRST THING YOU SEE

On successful connection, the z/OS system will display an initial screen that looks like the screenshot at right.

The initial screen is customized per installation. Your company's mainframe systems will have their own version of this. Regardless of how it looks, there will be a field where you can specify which mainframe environment you want to access.

In this setup, you enter "TSO userid" where "userid" is the mainframe userid you were given. You will not need to access any other environments on the mainframe.

```
IBM z/OS Standard Development/Test Image
Built on: 2024-03-21-16:35
System:      VS01
Workstation IP: 71.226.56.53
Date:       01/25/26
Time:       16:59:04

To logon to an Application type one of:
--> LOGON XXX (XXX is Vtam APPL)
--> TSO userid
--> CICSTS61
--> IMS15APL
--> CNM01

ENTER CHOICE ==> |
```

```
.
|
|
(ooo)
|
|
<<I>>
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DevSecOps
==CAN==
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|I|
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|MI|
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|CI|
|AI|
|NI|
|AI|  |□|
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/*****\ |AI|  |□□□|
(_____) |I-|  |□□□□|
||||||| |I-|  |□□□□|
```

GETTING INTO TSO

```
IBM z/OS Standard Development/Test Image
Built on: 2024-03-21-16:35
System:      VS01
Workstation IP: 71.226.56.53
Date:       01/26/26
Time:       06:51:27

To logon to an Application type one of:
--> LOGON XXX (XXX is Vtam APPL)
--> TSO userid
--> CICSTS61
--> IMS15APL
--> CNM01

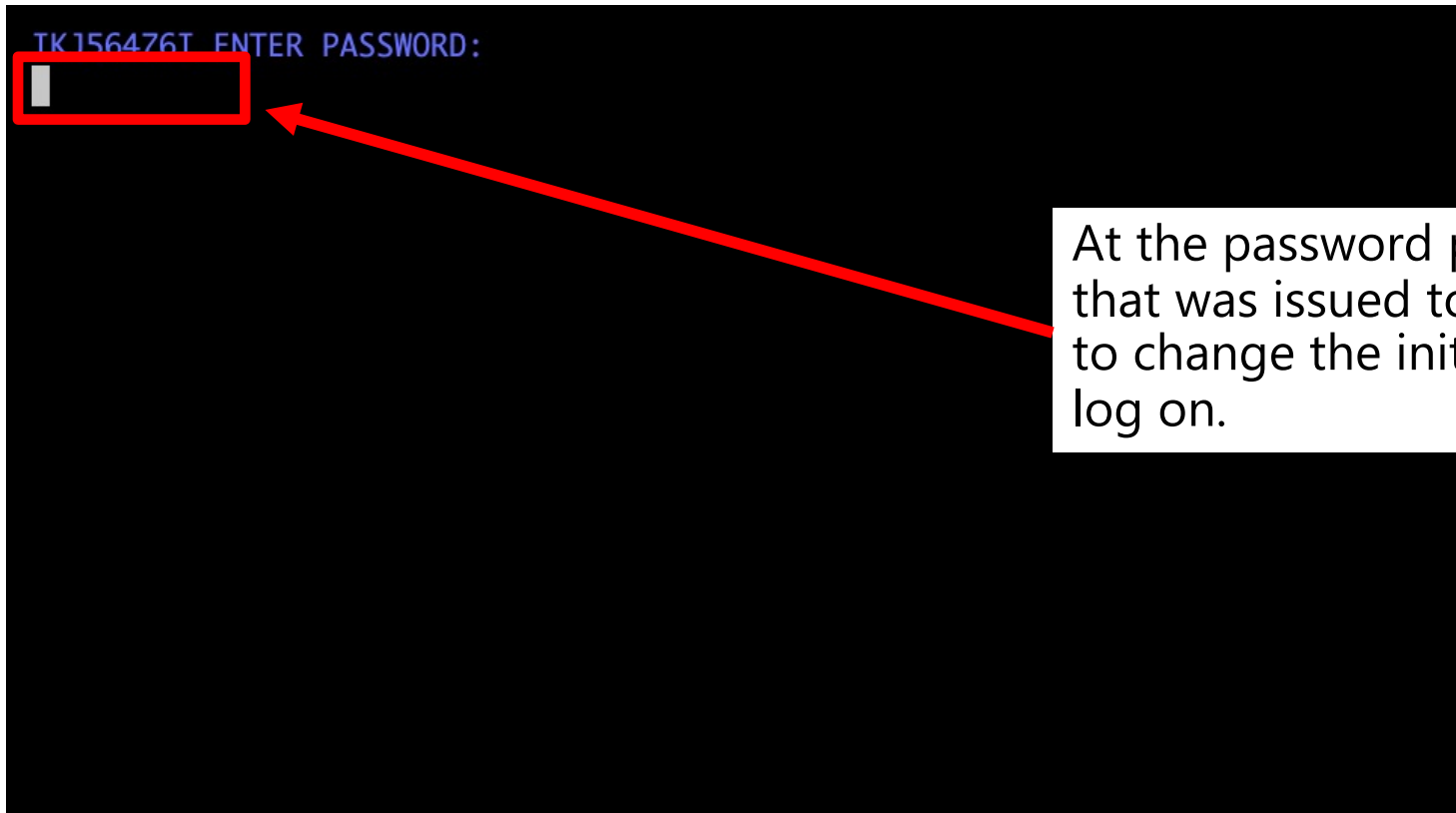
ENTER CHOICE ==> tso IBMUSER
```

```
.
|
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(ooo)
|
|
<<I>>
==IBM==
DevSecOps
==CAN==
<<->>
|I|
|B|
```

At the Command prompt, type **tso** and your userid and press Enter.

```
INI
|A|  |□|
|D|  |□□|
/*****\ |A|  |□□□|
(-----) ||-|| |□□□□|
||||||| ||-|| |□□□□|
```

PASSWORD PROMPT



At the password prompt, type the password that was issued to you. You may be prompted to change the initial password when you first log on.

TSOE LOGON SCREEN

```
----- TSO/E LOGON -----

Enter LOGON parameters below:

Userid    ==> IBMUSER

Procedure ==> DB13PROC

Acct Nbr  ==> ACCT001

Size      ==> 2096128

Perform   ==>

Command   ==> ispf

Enter an 'S' before each option desired below:
-New Password  -Nomail  -Nonotice  S -Reconnect  -OIDcard

PF1/PF13 ==> Help    PF3/PF15 ==> Logoff   PA1 ==> Attention   PA2 ==> Reshow
You may request specific help information by entering a '?' in any entry field
```

Group Ident ==>

z/OS installations define a default TSO configuration and default startup actions. Users can override some of those when they log on, if they need to work on specific tasks that call for it.

You do not need to change anything here and you can press Enter to proceed.

LOGON MESSAGES

```
ICH70001I IBMUSER  LAST ACCESS AT 15:41:04 ON SUNDAY, JANUARY 25, 2026
IKJ56455I IBMUSER LOGON IN PROGRESS AT 06:54:00 ON JANUARY 26, 2026
IKJ56951I NO BROADCAST MESSAGES
```

```
*****
* LOGON PROCS:                               *
* FOR PLAIN LOGON PROC USE PROC001          *
* FOR DB2 VERSION 13 USE DB13PROC           *
* FOR DB2 VERSION 12 USE DB12PROC           *
* FOR MQSERIES V9.3  USE MQPROC              *
*****
```

```
ispf
*** █
```

On successful logon the system will display any pending messages for your userid. These may include completion statuses of jobs that you left running the last time you logged off, broadcast messages, or the mainframe equivalent of "message of the day." Press Enter to dismiss the display.

ISPF PRIMARY OPTION MENU

```

Menu  Utilities  Compilers  Options  Status  Help
-----
ISPF Primary Option Menu

Option ==>
0  Settings      Terminal and user parameters      User ID . : IBMUSER
1  View          Display source data or listings    Time. . . : 11:06
2  Edit          Create or change source data       Terminal. : 3278
3  Utilities     Perform utility functions         Screen. . : 1
4  Foreground    Interactive language processing    Language. : ENGLISH
5  Batch         Submit job for language processing  Appl ID . : ISR
6  Command       Enter TSO commands                TSO logon : DB2PROC
7  Dialog Test   Perform dialog testing            TSO prefix: IBMUSER
9  IBM Products  IBM program development products   System ID : VS01
10 SCLM          SW Configuration Library Manager   MVS acct. : ACCT001
11 Workplace    ISPF Obj
12 z/OS System   z/OS sys
13 z/OS User     z/OS use

Enter X to Terminate

```

Most z/OS installations configure TSO to start ISPF automatically when a user logs on. ISPF stands for Interactive System Productivity Facility. It's a TSO application that wraps TSO commands in relatively user-friendly screens, called *panels*. ISPF is the basic user interface for developers and system administrators on z/OS systems.

As an infrastructure engineer, on those occasions when you must log onto z/OS directly, you will mainly use ISPF to do your work.

ACCESS THE USS COMMAND LINE

```
Menu Utilities Compilers Options Status Help
ISPf Primary Option Menu
Option ==> 6
0 Settings      Terminal and user parameters      User ID . : IBMUSER
1 View          Display source data or listings      Time. . . : 08:50
2 Edit          Create or change source data        Terminal. : 3278
3 Utilities     Perform utility functions          Screen. . : 1
4 Foreground    Interact
5 Batch         Submit j
6 Command       Enter TS
7 Dialog Test   Perform
9 IBM Products  IBM prog
10 SCLM         SW Configuration Library Manager    MVS acct. : ACCT001
11 Workplace    ISPF Object/Action Workplace         Release . : ISPF 8.1
12 z/OS System  z/OS system programmer applications
13 z/OS User    z/OS user applications

Enter X to Terminate using log/list defaults
```

To access the Unix System Services (USS) command line, navigate to the ISPF Command panel. By default this is option 6 on the Primary Option Menu, and most installations don't change it.

ACCESS THE USS COMMAND LINE

```
Menu List Mode Functions Utilities Help
ISP Command Shell
Enter TSO commands below:
==> omvs
Place cursor on choice and
=> omvs
=> D IPLINFO
=> ISRDDN
=> alloc da(qsam1) dsorg(ps) space(2,0) tracks lrecl(80) blksize(8000) recfm(f
=> alloc da(ex1.data) dsorg(ps) space(2,0) tracks lrecl(80) blksize(8000) recf
=> alloc da(ex1.data) dsorg(ps) space(2,0) tracks lrecl(80) +
=> alloc da(qsam1) new catalog space(1,1) trk dsorg(ps) recfm(fb) lrecl(80) bl
=> IND$FILE GET JCL(IN2OUT) ASCII CRLF
=> IND$FILE GET IBMUSER.JCL(IN2OUT) ASCII CRLF
=> ping 192.168.100.35
```

Enter the TSO command, omvs. (You could do this from the TSO ready prompt, too.) Press Enter.

ACCESS THE USS COMMAND LINE

```
IBM  
Licensed Material - Property of IBM  
5650-ZOS Copyright IBM Corp. 1993, 2017  
(C) Copyright Mortice Kern Systems, Inc., 1985, 1996.  
(C) Copyright Software Development Group, University of Waterloo, 1989.  
  
U.S. Government Users Restricted Rights -  
Use, duplication or disclosure restricted by  
GSA ADP Schedule Contract with IBM Corp.  
  
IBM is a registered trademark of the IBM Corp.  
  
IBMUUSER : /u/ibmuser : >
```

Enter the command, sh, to start the shell.

```
==> sh
```

ACCESS THE USS COMMAND LINE

```
IBMUSER : /u/ibmuser : > sh  
#
```

See what shell we're running.

```
=> echo $SHELL
```

ACCESS THE USS COMMAND LINE

```
IBMUSER : /u/ibmuser : > sh
```

```
# echo $SHELL
```

```
/bin/sh
```

```
#
```

Basic shell is running.



ACCESS THE USS COMMAND LINE

```
IBMUSER : /u/ibmuser : > sh
# echo $SHELL
/bin/sh
#
```

```
==>
```

```
uname -a
```

Check OS version.

ACCESS THE USS COMMAND LINE

```
IBMUSER : /u/ibmuser : > sh
# echo $SHELL
/bin/sh
# uname -a
0S/390 VS01 29.00 05 8562
#
```

It's z/OS (OS/390 in this case), not UNIX.
z/OS includes POSIX functionality, although
it is not 100% POSIX compliant.

ACCESS THE USS COMMAND LINE

```
IBMUSER : /u/ibmuser : > sh
# echo $SHELL
/bin/sh
# uname -a
OS/390 VS01 29.00 05 8562
# hostctl
hostctl: FSUM7351 not found
#
```

Type "exit" and press Enter to exit the shell.

==> exit

ACCESS THE USS COMMAND LINE

```
IBMUSER : /u/ibmuser : > sh
# echo $SHELL
/bin/sh
# uname -a
OS/390 VS01 29.00 05 8562
# hostctl
hostctl: FSUM7351 not found
# exit
IBMUSER : /u/ibmuser : >
```

Type "exit" and press Enter to exit OMVS.

⇒ exit

ACCESS THE USS COMMAND LINE

```
Menu List Mode Functions Utilities Help
ISPf Command Shell
Enter TS0 commands below:
==>

Place cursor on choice and press enter to Retrieve command

=> omvs
=> D IPLINFO
=> ISRDDN
=> alloc da(qsam1) dsorg(ps) space(2,0) tracks lrecl(80) blksize(8000) recfm(f
=> alloc da(ex1.data) dsorg(ps) space(2,0) tracks lrecl(80) blksize(8000) recf
=> alloc da(ex1.data) dsorg(ps) space(2,0) tracks lrecl(80) +
=> alloc da(qsam1) new catalog space(1,1) trk dsorg(ps) recfm(fb) lrecl(80) bl
=> IND$FILE GET JCL(IN2OUT) ASCII CRLF
=> IND$FILE GET IBMUSER.JCL(IN2OUT) ASCII CRLF
=> ping 192.168.100.35
```

Now you are back on the ISPF Command panel.

EXIT FROM ISPF

```
Menu Utilities Compilers Options Status Help
ISPf Primary Option Menu
Option ==> x
0 Settings      Terminal and user parameters      User ID . : IBMUSER
1 View          Display source data or listings      Time. . . : 06:54
2 Edit          Create or change source data      Terminal. : 3278
3 Utilities      Perform
4 Foreground     Interact
5 Batch          Submit j
6 Command        Enter TS
7 Dialog Test    Perform
9 IBM Products   IBM program development products      System ID : VS01
10 SCLM          SW Configuration Library Manager      MVS acct. : ACCT001
11 Workplace     ISPF Object/Action Workplace          Release . : ISPF 8.1
12 z/OS System   z/OS system programmer applications
13 z/OS User     z/OS user applications

Enter X to Terminate using log/list defaults
```

To log off from z/OS, the first step is to exit from ISPF. Type "x" in the Option field on the Primary Option Menu, or type "=x" in the Command field of any ISPF panel, and press Enter.

SPECIFY DISPOSITION OF LOG DATA SET

```
Specify Disposition of Log Data Set

Command ==> _____

Log Data Set (TRMISER SPELOG1.LIST) Disposition:
Process Option . . . . 2
1. Print data set and delete
2. Delete data set without printing
3. Keep data set - Same
   (allocate same data set in next session)
4. Keep data set - New
   (allocate new data set in next session)

Batch SYSOUT class . . _____
Local printer ID or
writer-name . . . . . _____
Local SYSOUT class . . _____

List Data Set Options not available

Press ENTER key to complete ISPF termination.
Enter END command to return to the primary option menu.

Job statement information: (Required for system printer)
==> _____
==> _____
==> _____
==> _____
```

ISPF keeps a log of your actions during your session. There is no need to keep or print the log. Choose option 2 to delete it.

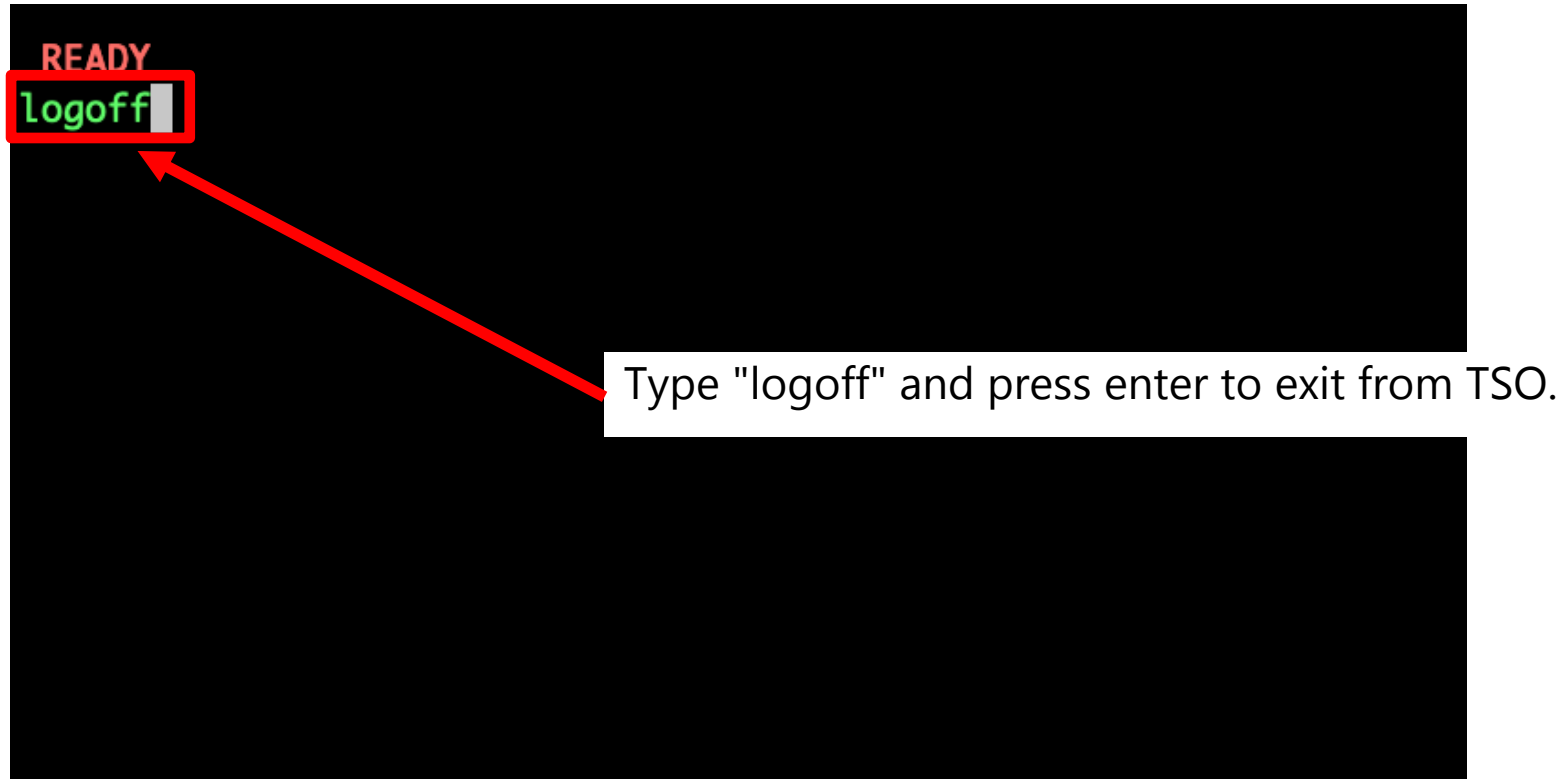
TSO READY PROMPT

```
IBMUER.SPFLOG1.LIST has been deleted.  
READY
```

Exiting ISPF takes you to the TSO ready prompt.

You can enter basic TSO commands from here, but most people use the ISPF Command panel rather than the TSO ready prompt.

EXIT FROM TSO



EXIT FROM TSO

IBM z/OS Standard Development/Test Image

Built on: 2024-03-21-16:35

System: VS01

Workstation IP: 71.226.56.53

Date: 01/26/26

Time: 06:56:36

.
|
|
(ooo)
|
|
<<I>>
==IBM==
DevSecOps

To logon to an Application type one of:

--> LOGON XXX (XXX is Vtam APPL)

--> TSO userid

--> CICSTS61

--> IMS15APL

--> CNM01

ENTER CHOICE ==> logoff

You are still connected to the mainframe at this point. You could access some other application besides TSO from here.

There may be times when you do that in the normal course of your work, but you will not need to do so for this course.

To disconnect from the mainframe, type "logoff" at the prompt and press Enter.

IAI I□I
IDI I□□I
/*****\ IAI I□□□I
(-----) II-II I□□□□I
IIIIIIII II-II I□□□□I

LAB 1: CONNECT TO ZOS VIA TERMINAL EMULATOR

Follow the instructions for Lab 1 under labs/labs.md in the course repo. This will look similar to the sequence of steps we've just walked through. Don't worry if you need help.

ANSIBLE CONTROL NODE

Set the mainframe userid that was assigned to you in your inventory.yml file.

```
"inventory.yml" 10L, 256B
all:
  children:
    zos:
      hosts:
        mainframe:
          ansible_host: 67.217.62.83
      vars:
        ansible_user: IBMUSER
        ansible_ssh_private_key_file: id_rsa
        ansible_python_interpreter: /usr/lpp/IBM/cyp/v3r11/pyz/bin/python
```

LAB 2: USE ANSIBLE TO PING THE Z/OS INSTANCE

Follow the instructions for Lab 2 under labs/labs.md in the course repo. This will look similar to the sequence of steps we've just walked through. Don't worry if you need help.

FINDING HELP WITH ZOS AND ANSIBLE



Documentation



Community

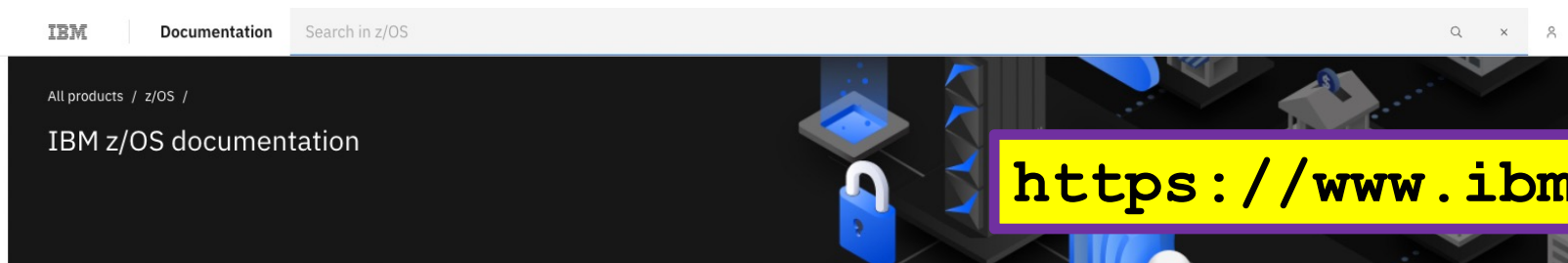


AI Assist

Source Code

```
32 class ActionModule(ActionBase):
33     def run(self, tmp=None, task_vars=None):
34         """ handler for file transfer operations """
35         self._supports_async = True
36
37         if task_vars is None:
38             task_vars = {}
39
40         result = super(ActionModule, self).run(tmp, task_vars)
41
42         if result.get("skipped"):
43             return result
44
45         module_args = self._task.args.copy()
46
47         use_template = _process_boolean(module_args.get("use_template"))
48         remote_src = module_args.get("remote_src")
49         if use_template and remote_src:
50             result.update(dict(
51                 failed=True,
52                 changed=False,
53                 msg="Use of Jinja2 templates is only valid for local files. remote_src is set"
54             ))
55         return result
```

IBM DOCUMENTATION



<https://www.ibm.com/docs/en/zos>

z/OS 3.1.0	z/OS 2.5.0	z/OS 2.4.0	z/OS 2.3.0
Version 3.1.0	Version 2.5.0	Version 2.4.0	
z/OS 2.2.0	z/OS 2.1.0		
Version 2.2.0	Version 2.1.0		

IBM has always been very good about documentation. Their online documentation is comprehensive, accurate, and up to date. It will be your primary source of help with using z/OS and other IBM products.

Resources

IBM Support Portal

Get assistance for the IBM products, services and software you own.

IBM Fix Central

Provides fixes and updates for your system's software, hardware, and operating system.

IBM Passport Advantage

Provides you with secure access to software downloads and media.

IBM DOCUMENTATION

In the case of IBM products, RTFM (Read the Fine Manual) is sound advice. It is not always given in a respectful or supportive way, but it is sound advice nonetheless.

One of the most useful things you can do is to spend some time exploring the online IBM documentation for z/OS and products you will be using on that platform, such as JCL, z/OSMF, z/OS Provisioning, and SA z/OS.

Let's see how the documentation is organized and structured, and how to get the most value from it.



IBM DOCUMENTATION – URI STRUCTURE

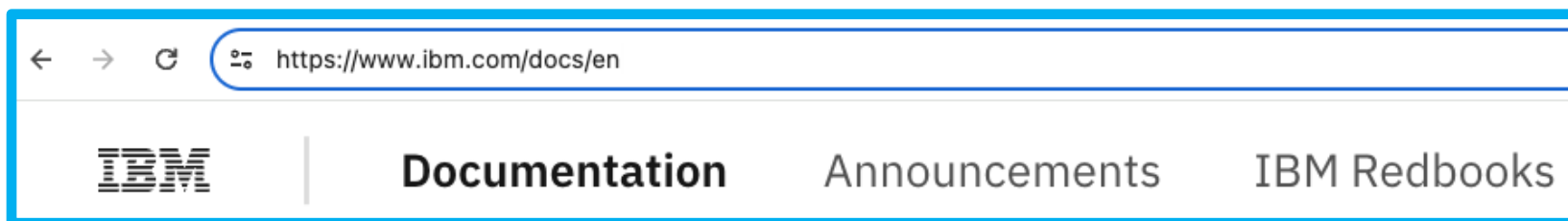
If you're not sure where to start looking for information, you may be able to make a pretty good guess if you understand how IBM structures the URIs for documentation. Here's the top level documentation URI:

`https://ibm.com/docs`

The next segment in the path is the **ISO 639 language code**. The server will default this based on the HTTP Content-Language and/or Accept-Language headers in the request, or on the presumed location of your IP address. In my case, it defaults to English, like this:

`https://ibm.com/docs/en`

Here's the top portion of the resulting document:

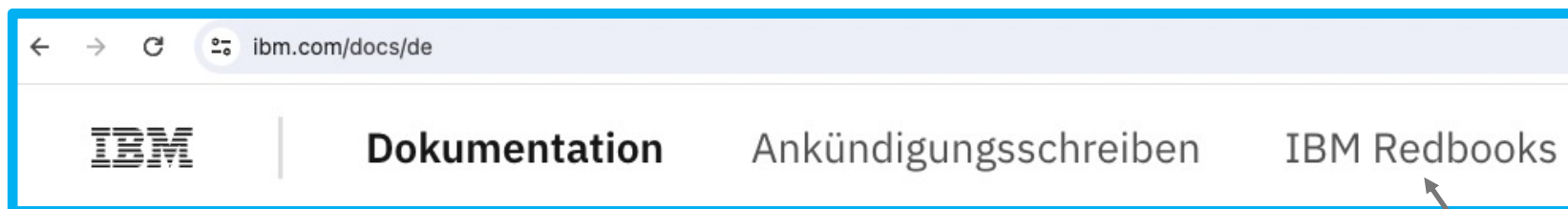


IBM DOCUMENTATION – ISO 639 LANGUAGE CODE

We can change the language code in the URI and reload the page to see the information in other languages, if the particular page is available in the language we specify. Otherwise the default is English.

Here's the same page in German, when we change the language code to **de**.

<https://ibm.com/docs/de>



To get the page in Chinese, we can change the language code to **zh**.

<https://ibm.com/docs/zh>



*Trademarked
product names are
not translated*

IBM DOCUMENTATION – PRODUCT NAME

After the language code, the next path segment is a product name. This is not necessarily the official IBM product name, since those often contain slashes or multiple words or a particular combination of upper- and lowercase letters. This will be an abbreviated product name that makes sense as part of a URI. For z/OS, it's **zos** – one of the more guessable examples.

<https://ibm.com/docs/en/zos>

IBM Documentation Search in z/OS

All products / z/OS /

IBM z/OS documentation

z/OS 3.1.0	z/OS 2.5.0	z/OS 2.4.0	z/OS 2.3.0
Version 3.1.0	Version 2.5.0	Version 2.4.0	
z/OS 2.2.0	z/OS 2.1.0		
Version 2.2.0	Version 2.1.0		

The format of this page is typical of the main page of IBM documentation for a given product. The upper portion contains links to each version of the product for which documentation is available online. IBM rolls older documentation off the Internet as new versions of the product come onto the market. In this case we have the current version of z/OS plus the five most recent previous versions.

IBM DOCUMENTATION – PRODUCT VERSION

The next section of the page contains several links under the heading, **Getting Started**. We'd like to call your attention to one of them, **IBM z/OS Adobe Indexed PDF Collection**. You can obtain nearly all the IBM documentation in the form of PDF files, which you can download free of charge. You may find this format easier to work with for documents you refer to frequently, or that you want to peruse offline.

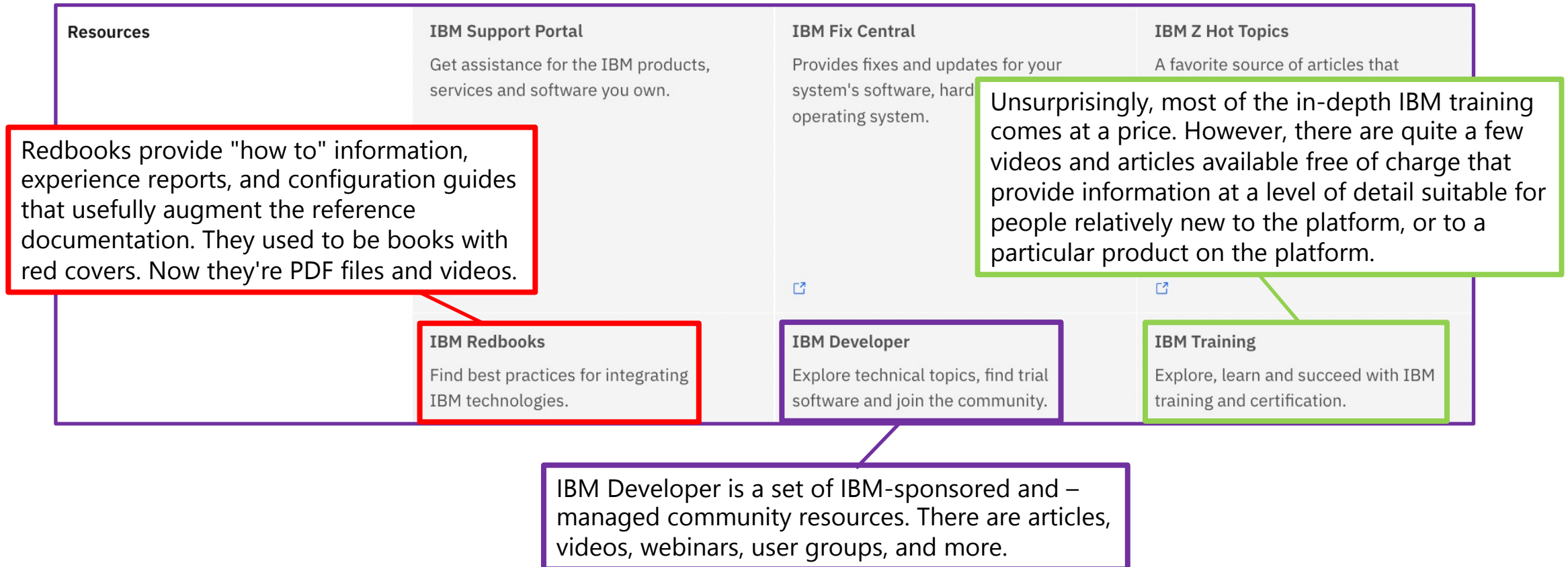
<https://ibm.com/docs/en/zos/2.5.0>

Getting started	IBM z/OS Adobe Indexed PDF Collection (1.1GB)	What's New in z/OS 2.5.0	z/OS home page
	z/OS 2.5.0 Program Directory	z/OS Introduction and Release Guide	z/OS Release Upgrade Reference Summary

IBM DOCUMENTATION – PRODUCT VERSION

Still farther down the same page, under **Resources**, there are several links to additional documentation and resources.

<https://ibm.com/docs/en/zos/2.5.0>



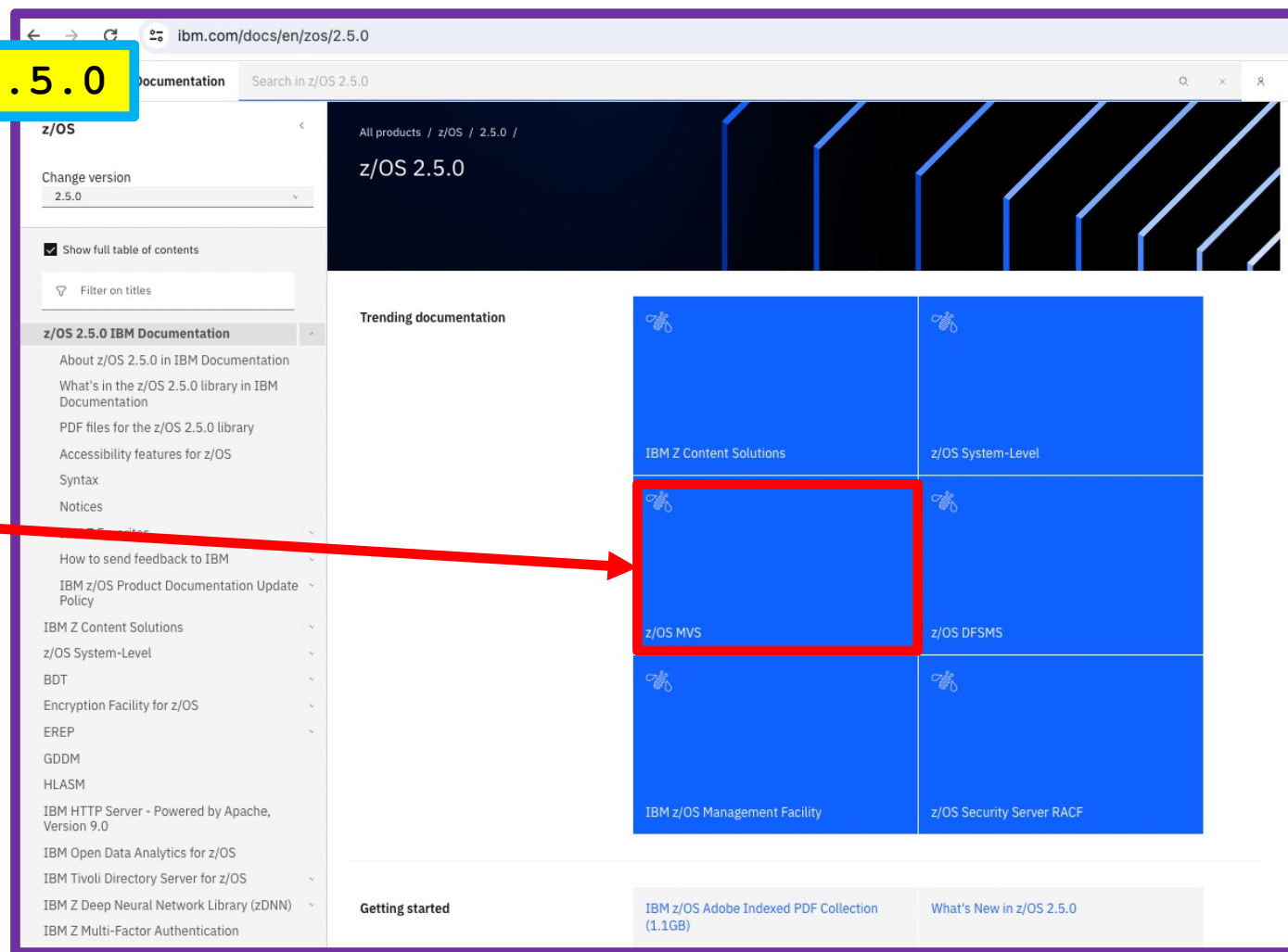
IBM DOCUMENTATION – PRODUCT VERSION

If we follow the link for version 2.5.0, we reach the top documentation page for that version of z/OS.

<https://ibm.com/docs/en/zos/2.5.0>

Most of the information we will be interested in for the bootcamp falls under the heading of "z/OS MVS".

MVS (Multiple Virtual Storage) was the name of an ancestor of z/OS. The term MVS refers to elements of z/OS that were part of that operating system. They are still core components of z/OS today.



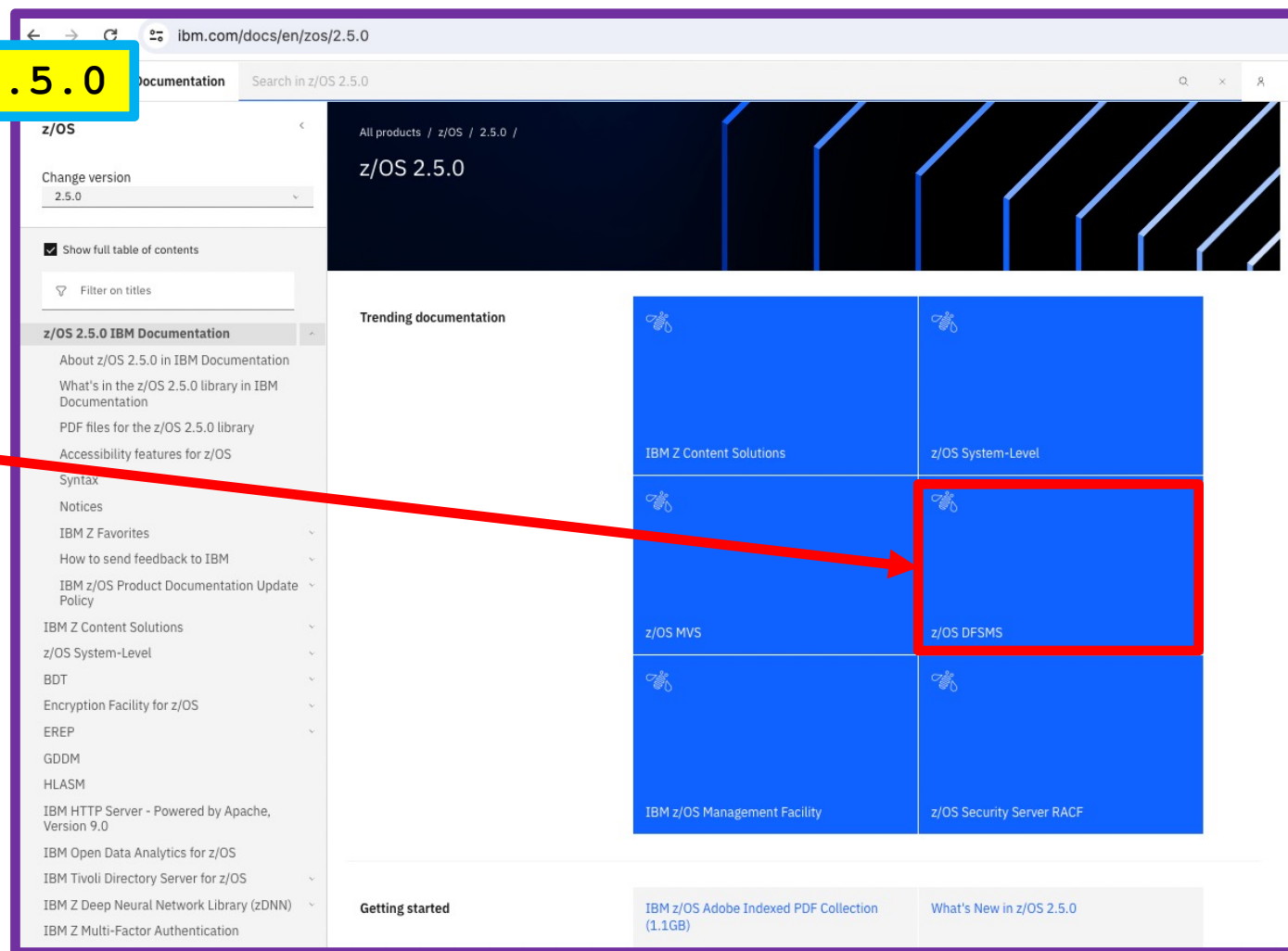
IBM DOCUMENTATION – PRODUCT VERSION

If we follow the link for version 2.5.0, we reach the top documentation page for that version of z/OS.

<https://ibm.com/docs/en/zos/2.5.0>

We'll also use the documentation under z/OS DFSMS during the bootcamp.

Let's follow that link now and see where it leads us.



IBM DOCUMENTATION – TOPIC

At this point, the server doesn't add anything to the path, but introduces a query string with the keyword, **topic**.

<https://ibm.com/docs/en/zos/2.5.0?topic=zos-dfsms>

We aren't going to drill into this topic here and now. We want to show you how to navigate the different versions of a topic. All the reference documentation has this sort of navigation.

First, the "newer version" pop-up is present because we navigated to this page from a z/OS 2.5.0 page, which is not the most recent version for which online documentation is available. If we follow the **View latest** link, we'll get to the top-level z/OS documentation page for the latest version of z/OS.

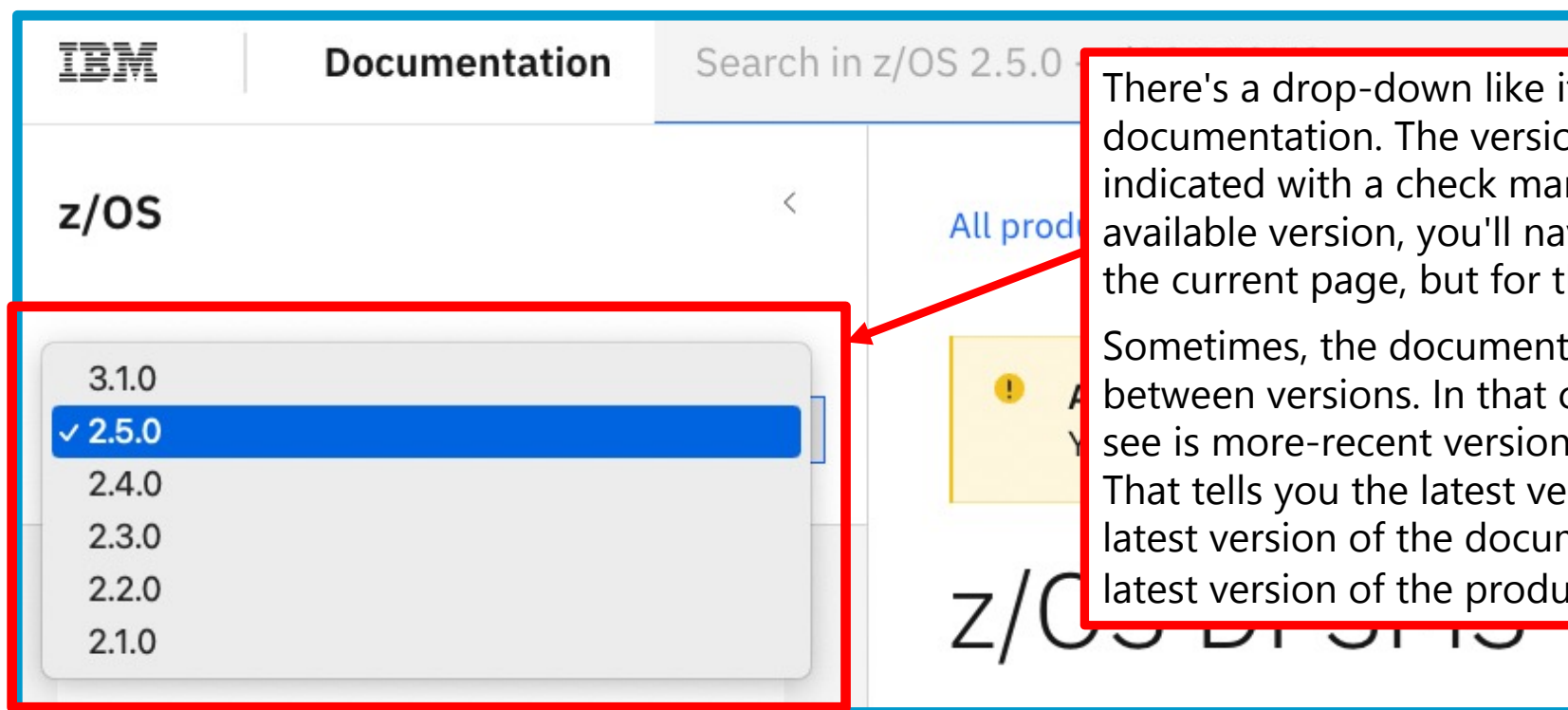
But that is not what people usually want. They usually want to go to the latest (or a different) version of the *same topic*. The next slide explains how to do that.

The screenshot displays the IBM Documentation interface for the topic "z/OS DFSMS". The top navigation bar includes the IBM logo, the word "Documentation", and a search bar containing "Search in z/OS 2.5.0 - z/OS DFSMS". On the left side, there is a sidebar with "z/OS" and a "Change version" dropdown menu currently set to "2.5.0". Below this is a checkbox labeled "Show full table of contents" which is checked. The main content area shows a breadcrumb trail: "All products / z/OS / 2.5.0 /". A yellow notification banner is present, stating: "A newer version of this product documentation is available. You are viewing an older version." A red box highlights the "View latest" link within this banner. To the right of the banner, there is a "Was this topic helpful?" section with thumbs up and down icons. The main title "z/OS DFSMS" is displayed at the bottom of the content area.

IBM DOCUMENTATION – TOPIC

One way to reach the same topic, but for a different version of the product, is to change the product version in the URI. The other way is to look at the versions listed under the drop-down labeled, "Change version".

<https://ibm.com/docs/en/zos/2.5.0?topic=zos-dfsms>



There's a drop-down like it on every page of reference documentation. The version you're looking at is indicated with a check mark. If you select any other available version, you'll navigate to the same topic as the current page, but for the selected version.

Sometimes, the documentation does not change between versions. In that case, the convention you will see is more-recent version numbers appear in italics. That tells you the latest version in normal typeface is the latest version of the documentation, even if not the latest version of the product itself.

IBM DOCUMENTATION – SEARCHING

There's a search box on every page of reference documentation. If we type in **jcl** and press Enter, we'll get a link to every page of documentation that mentions "jcl". Sometimes we get lucky, but usually we get a very large number of results that are not relevant. It can be quicker to use a generic Internet search engine like DuckDuckGo.

The screenshot shows the IBM Documentation search interface. The top navigation bar includes the IBM logo, the word "Documentation", and a search bar containing the text "jcl". On the left side, under the heading "Technical documentation", there is a list of categories with radio buttons: IBM Documentation (selected), IBM Developer, Support Technotes, IBM Redbooks, IBM Announcement letters, IBM Sales manuals, IBM Training, and IBM API Hub. At the bottom of this list is a link to "IBM.com". The main content area displays "1,144 results for 'jcl' in z/OS DFSMS." Below this, there is a "Change search scope:" dropdown menu currently set to "z/OS DFSMS". Two search results are visible, each preceded by a document icon and the text "JCL". The first result is titled "z/OS / 2.5.0 / z/OS DFSMS / z/OS DFSMS Using Data Sets" and includes a brief description and a "1 Minute" read time. The second result is titled "z/OS / 2.5.0 / z/OS DFSMS / z/OS DFSMSrmm Implementation and Customization Guide" and also includes a brief description and a "1 Minute" read time.

IBM | Documentation | jcl

Technical documentation

- ☒ IBM Documentation
- ☐ IBM Developer
- ☐ Support Technotes
- ☐ IBM Redbooks
- ☐ IBM Announcement letters
- ☐ IBM Sales manuals
- ☐ IBM Training
- ☐ IBM API Hub

[IBM.com](#)

1,144 results for "jcl" in z/OS DFSMS.

Change search scope: z/OS DFSMS

JCL

z/OS / 2.5.0 / z/OS DFSMS / z/OS DFSMS Using Data Sets

All data sets can be defined directly through **JCL**. Related reading: For information about using **JCL**, see z/OS MVS **JCL** Reference and z/OS MVS **JCL** User's Guide. Parent topic: Allocating data sets ...

1 Minute

JCL for EDGUTIL

z/OS / 2.5.0 / z/OS DFSMS / z/OS DFSMSrmm Implementation and Customization Guide

This topic provides **JCL** for creating the DFSMSrmm control data set, verifying the contents of the control data set, updating the control data set, and mending the control data set. The MASTER DD statement is required.,**JCL** for creating the control data set **JCL** for updating the control data set **JCL** for verifying the contents of the...

1 Minute

LAB 3: LOCATE IBM DOCUMENTATION ONLINE

Follow the instructions for Lab 3 under labs/labs.md in the course repo.

COMMUNITY

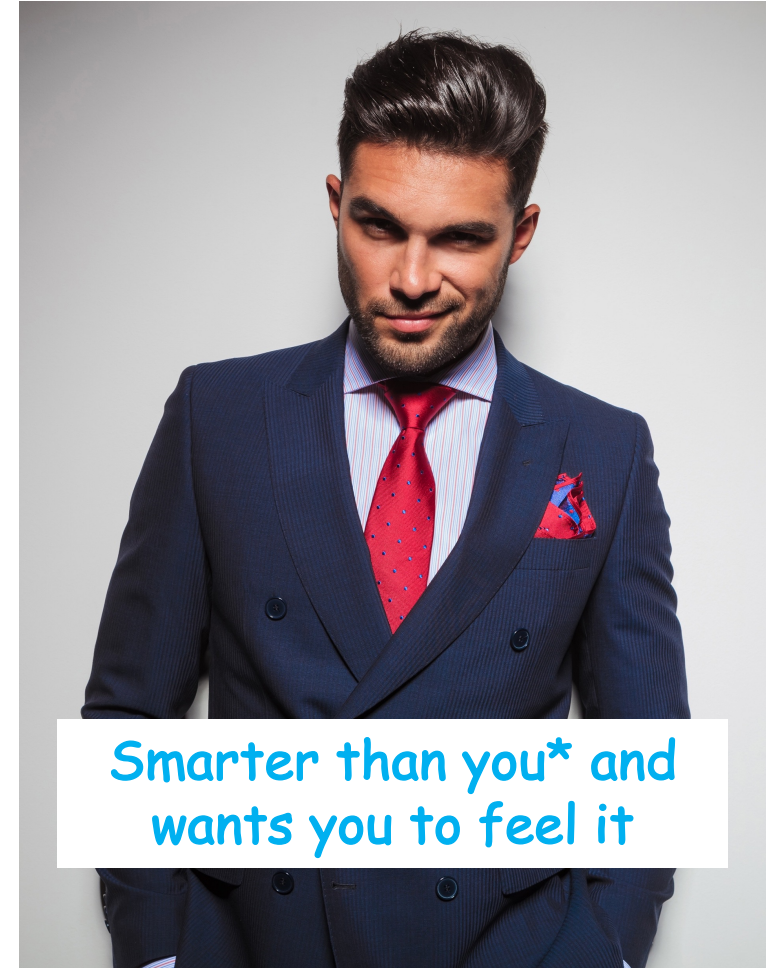
There are three kinds of community resources where you might seek help:

- IBM-sponsored and -managed community resources
- Third-party sites that offer explanatory documentation and tutorials
- Discussion boards and mailing lists

IBM-sponsored resources tend to be open and respectful of newcomers to IBM technologies. The company does not want to alienate new users.

Third-party sites that appear well-organized and present information in an authoritative style are often out of date or just plain wrong in what they say. Exercise caution when using this type of resource.

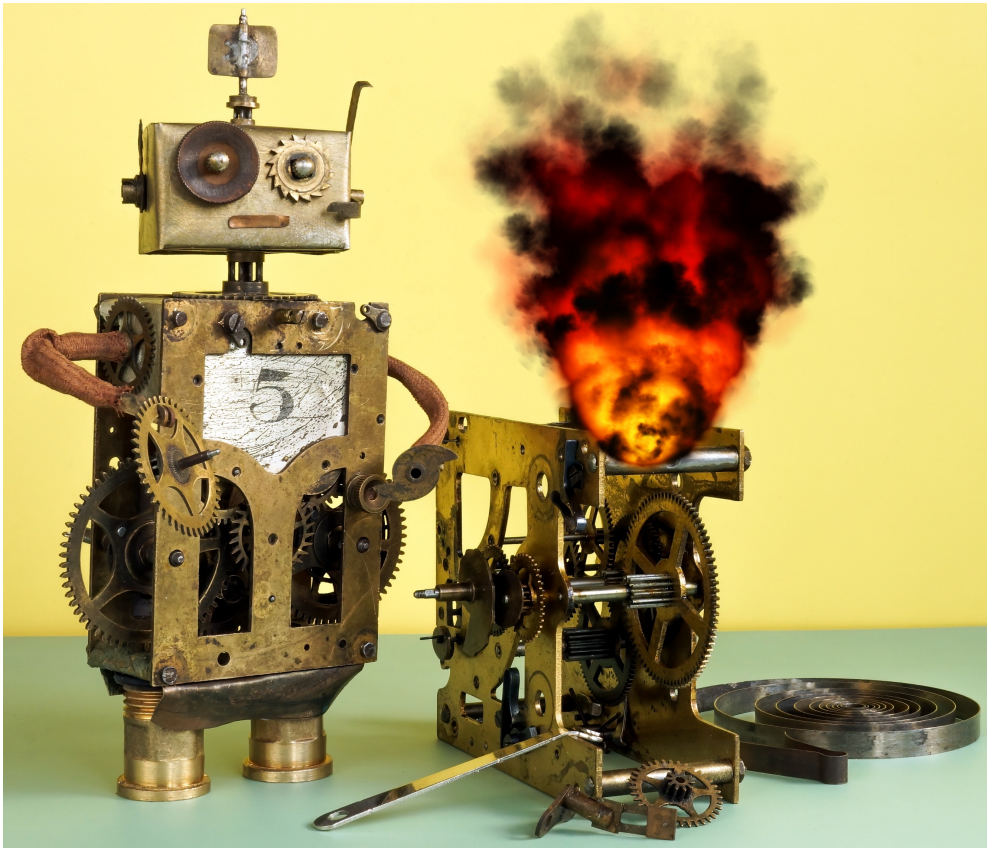
There are issues with volunteer-run discussion boards focusing on mainframe technologies. **First**, the most prolific people answering questions seem to prefer making others feel stupid over helping them learn. Getting to a clear answer is an unpleasant exercise. **Second**, some of the most helpful are retirees with no access to a live z/OS system. They rely on memory and documentation to answer questions. **Third**, many people describe a workaround that got them through a similar problem in the past, but it is not necessarily a reliable or general solution. They lack true in-depth knowledge of the system. It may be risky to copy what they did.



Smarter than you* and
wants you to feel it

* maybe

AI ASSIST



Artificial Intelligence tools have become popular in software development. While they may be helpful, it can be risky to depend on them too heavily.

It's useful to remember these tools are not really "intelligent," despite the name.

They are based on Large Language Models (LLMs) trained by examining billions of lines of code accessible on the Internet and in code bases they have been authorized to process.

The general level of quality of software is not so great. AI tools can't make judgments about that. They favor patterns that occur frequently, regardless of quality.

As a result, AI-based coding assistants often provide suggestions that are suboptimal.

The state of the art *circa* 2026 is that knowledgeable programmers can get value from an AI-based assistant because they can reject or modify suggestions the assistant makes.

Novices lack the experience to do that, and they tend to produce questionable results when they use an AI assistant.

This bootcamp is aimed at people who are new to z/OS and related technologies. Therefore, we recommend avoiding AI assistants until you have gained enough experience to understand the quality of the suggestions they offer.

WHAT YOU KNOW AND CAN DO

- You are a committer for the course repository on Github. You have successfully pushed a text file to the repository that contains your name.
- You have a working IBM 3270 terminal emulation program installed on your local system or accessible to you over the network.
- You have tested your credentials for the z/OS lab environment and you can sign on and off that system.
- You have a general idea of how IBM documentation is organized and structured.

WHAT'S NEXT

- About IBM mainframes